Homework 2

Asset Allocation with Stock Portfolios and Bills

1. (10%) Please derive Equation (7.13) by maximizing the slope, $\frac{\mu_p - r_f}{\sigma_p}$, of CAL. See footnote 5 for the solution procedure.

2. (90%) You can find the dataset containing the annual returns of "Small/Value" and "Small/Growth" portfolios at the Professor Kenneth French's Web site http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

(Go to Kenneth French's Web site, find **6 Portfolios Formed on Size and Book-to-Market (2 x 3)**; Download the CSV file **(6_Portfolios_2x3.CSV)** and use **(Average Equal Weighted Returns -- Annual)**).

Copy the 20 years data (from year 2006 to 2025) into a new spreadsheet. Next analyze the risk-return tradeoff that would have characterized portfolios constructed from "Small/Value" and "Small/Growth" over this period of time.

(10%) (1) What were the average rate of return, standard deviation, skewness, and kurtosis of their annual returns?

(10%) (2) What were the covariance and the correlation coefficient of their annual returns?

(10%) (3) Draw the portfolio opportunity set of "Small/Value" and "Small/Growth" portfolios. Please consider weights in "Small/Growth" portfolio starting at -2 and incrementing by .10 up to a weight of 2.

(10%) (4) Redo (3) by setting correlation coefficients to -1 and +1. Please plot the expected returns and standard deviations on the same chart in problem (3)

Use data from year 2006 to 2025.

(10%) (5) What was the average return and standard deviation of the minimum-variance combination of "Small/Value" and "Small/Growth" portfolios?

Assume T-bill has annual return 1.6375% and **standard deviation 0**.

(10%) (6) Find the optimal risky portfolio (P) and its expected return and standard deviation.

(10%) (7) Find the slope of the CAL generated by T-bill and portfolio P. Draw the new portfolio opportunity set and plot it on the same chart in problem (3).

Assume the utility function of the textbook $U = E(r) - \frac{1}{2} A\sigma^2$.

(10%) (8) If an investor has a coefficient of risk aversion A=4, calculate the composition of the complete portfolio of T-bill, "Small/Value" and "Small/Growth" portfolios. Please plot the expected return and standard deviation of this complete portfolio as well as the indifference curve on the same chart in problem (3).

(10%) (9) If another investor has a coefficient of risk aversion A=1, calculate the composition of the complete portfolio of T-bill, "Small/Value" and "Small/Growth" portfolios. Please plot the expected return and standard deviation of this complete portfolio as well as the indifference curve on the same chart in problem (3).

Useful Excel Functions:

AVERAGE
STDEV
SKEW
KURT
COVAR